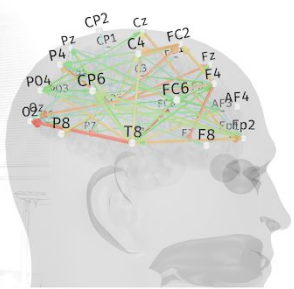
How do we educate? Research?

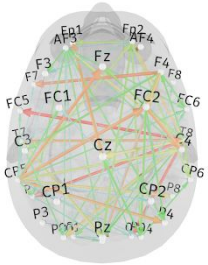
What are the objectives?

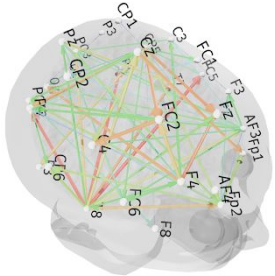
- Prevention?
- Critical thinking?
- Transparency?
- Academic curiosity?
- Skills development?
- Employability?
- Adding new knowledge?

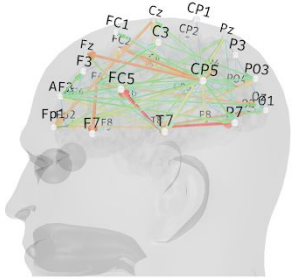
Problem 2: assessing the risks — and opportunities











***
more significant

p values

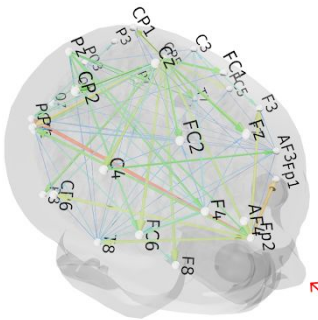
*
less significant

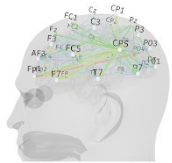
Your Brain on ChatGPT

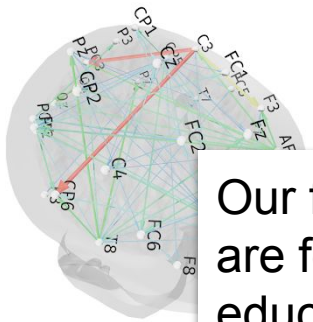
higher dDTF

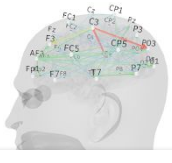
EEG Alpha Band

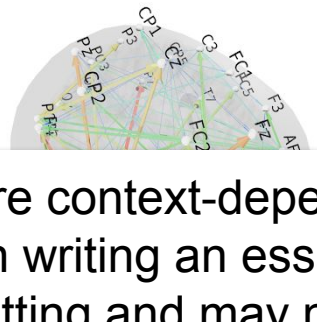
lower dDTF

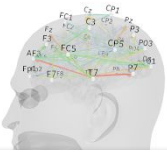




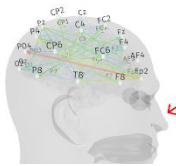


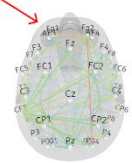


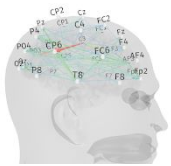




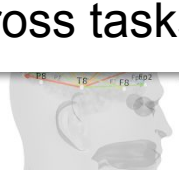
LLM

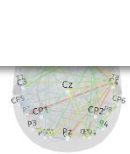












Our findings are context-dependent and are focused on writing an essay in an educational setting and may not generalize across tasks.

Nataliya Kosmyna & Eugene Hauptmann. Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task, MIT, 2025: https://www.brainonllm.com/

“You may just want to get AI guidance but it will often just give you the answer instead. this short-circuits the ***(sometimes unpleasant) mental effort that creates learning.****”*

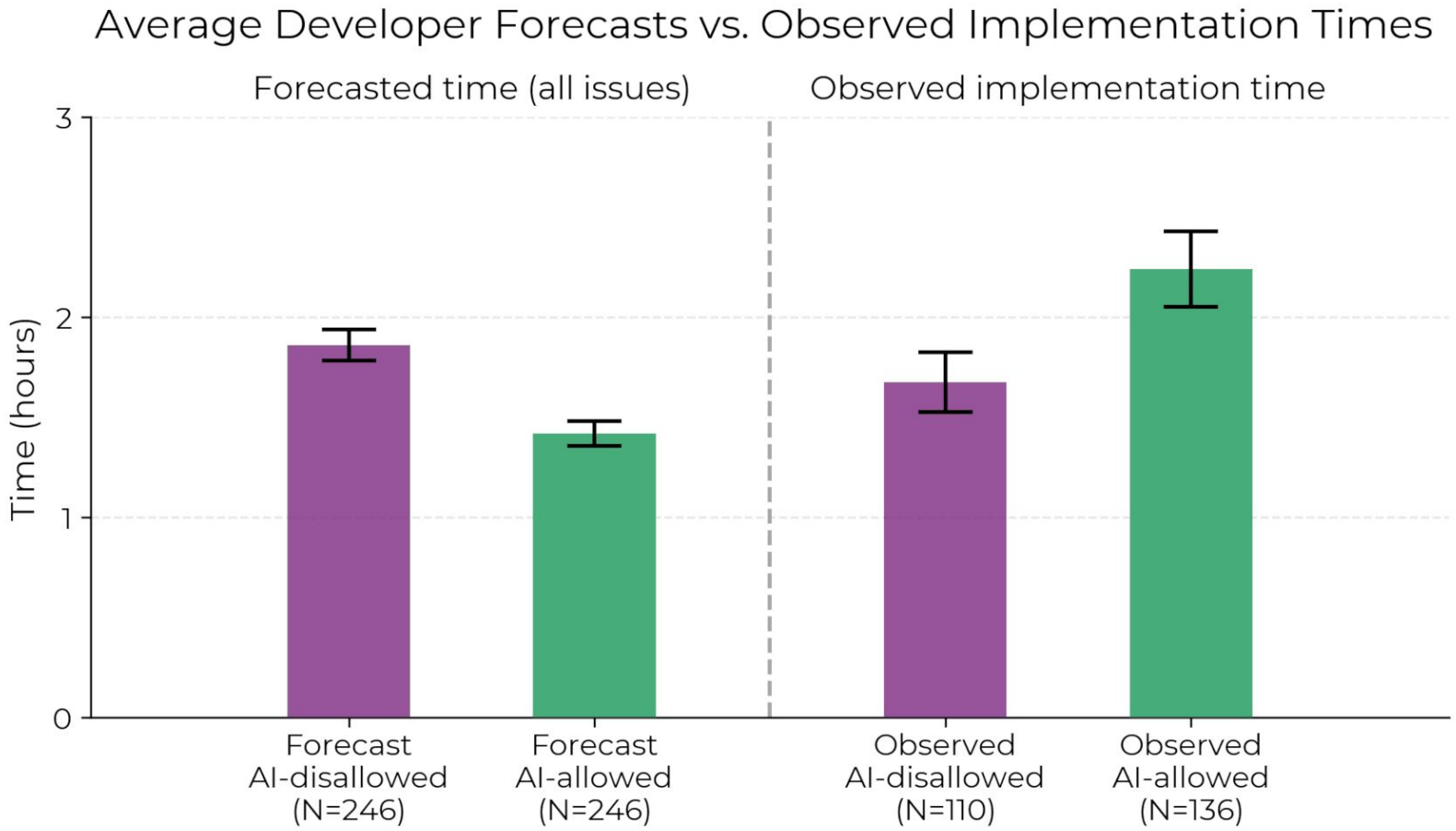
“Even though students ***thought they learned a lot*** *from ChatGPT's help, they* ***actually scored 17% worse****.”*

Mollick, Ethan. Against "Brain Damage", July 7 2025, https://www.oneusefulthing.org/p/against-brain-damage

*“AI adoption **raises homework scores by 18%** and reduces completion time by 30%, **but lowers monthly exam scores by 20%** within six months.*

The losses are largest in social science subjects, followed by STEM and languages, and are especially large for junior students, high-achieving students, and boys

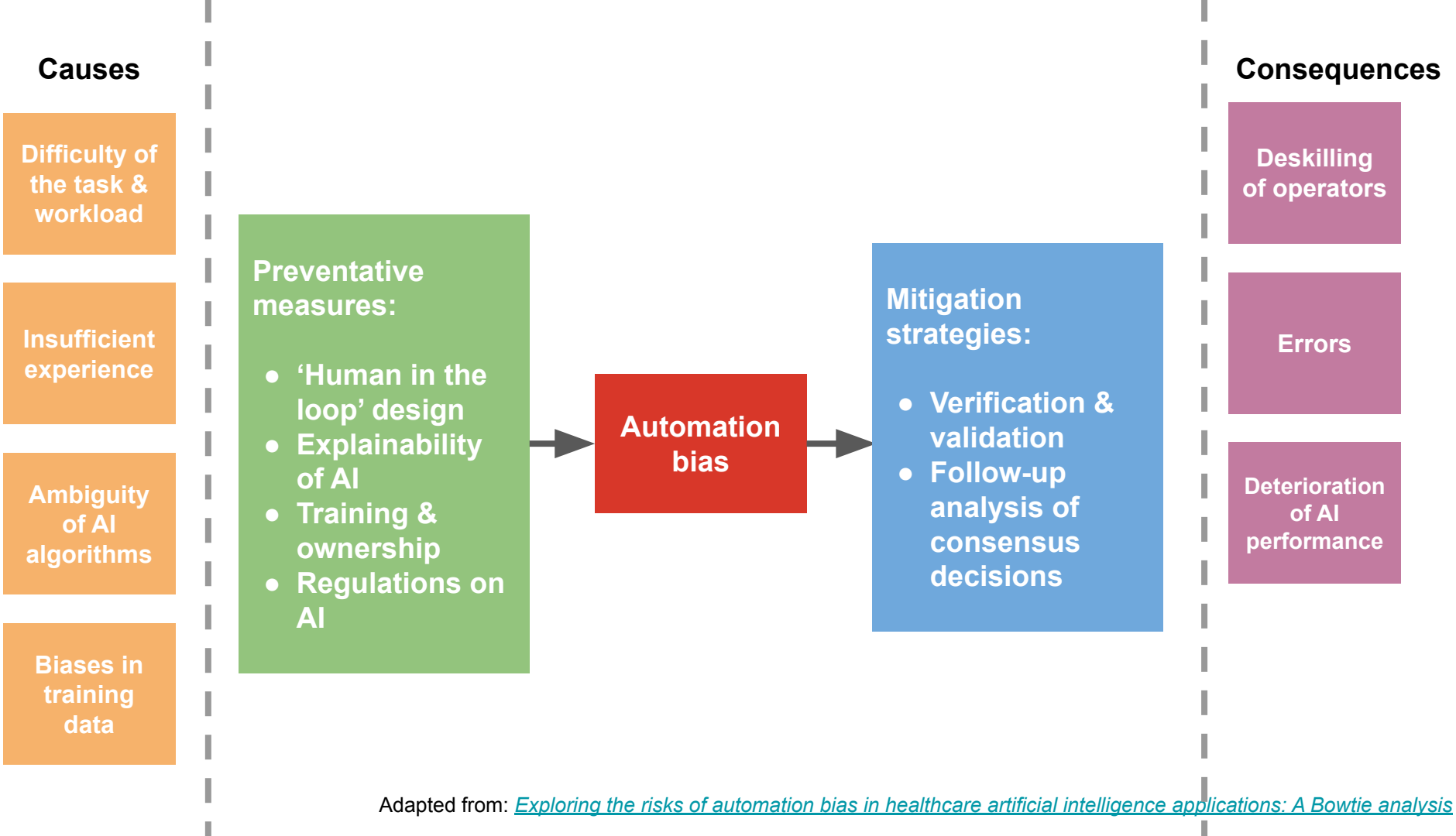
Strömberg, Wu & Lei. The Generative AI Learning Penalty: Evidence from Chinese Secondary Education, 2026, https://cepr.org/publications/dp21577



Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity (2025)
https://metr.org/blog/2025-07-10-early-2025-ai-experienced-os-dev-study/

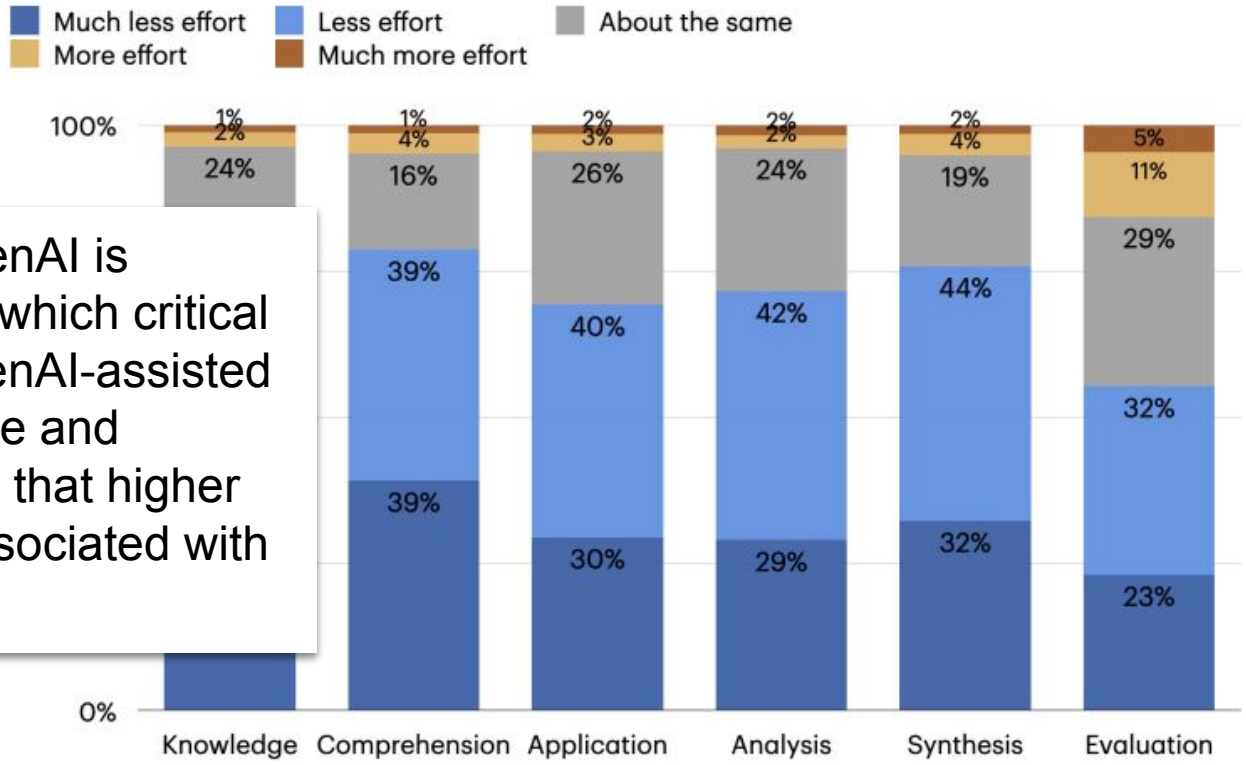
We do *not* provide evidence that:	Clarification
AI systems do not currently speed up many or most software developers	We do not claim that our developers or repositories represent a majority or plurality of software development work
AI systems do not speed up individuals or groups in domains other than software development	We only study software development
AI systems in the near future will not speed up developers in our exact setting	Progress is difficult to predict, and there has been substantial AI progress over the past five years [2]
There are not ways of using existing AI systems more effectively to achieve positive speedup in our exact setting	Cursor does not sample many tokens from LLMs, it may not use optimal prompting/scaffolding, and domain/repository-specific training/finetuning/few-shot learning could yield positive speedup

https://metr.org/blog/2025-07-10-early-2025-ai-experienced-os-dev-study/



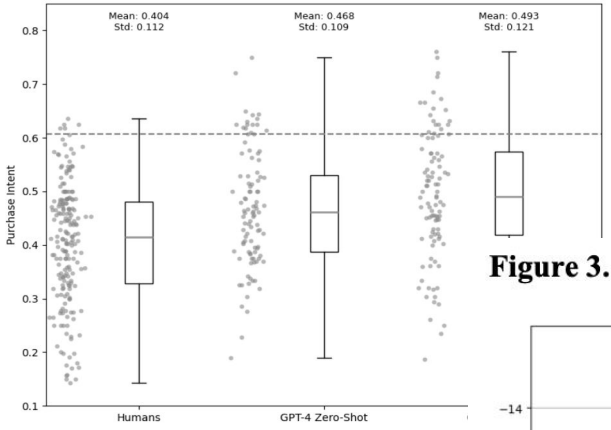
Adapted from: *Exploring the risks of automation bias in healthcare artificial intelligence applications: A Bowtie analysis*

“A user’s confidence in GenAI is predictive of the extent to which critical thinking is exercised in GenAI-assisted tasks. Both our quantitative and qualitative results suggest that higher confidence in GenAI is associated with less critical thinking”



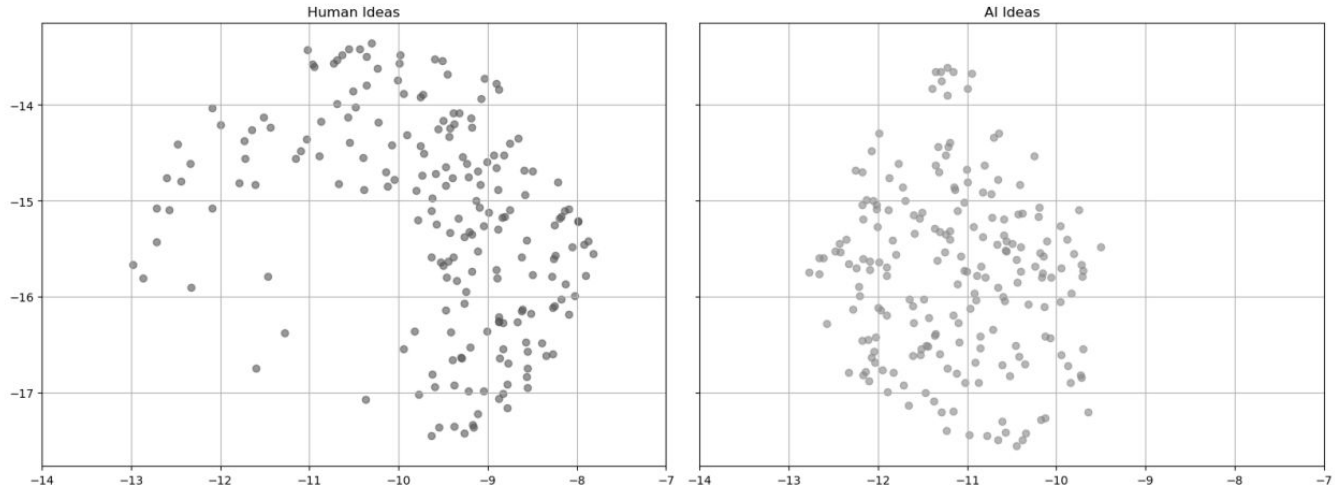
Distribution of perceived effort (%) in cognitive activities (based on Bloom’s taxonomy) when using a GenAI tool compared to not using one. From [The Impact of Generative AI on Critical Thinking: Self-Reported Reductions in Cognitive Effort and Confidence Effects From a Survey of Knowledge Workers - Microsoft Research]

Figure 1. Distribution of Idea Quality for Three Sets of Ideas



Notes. Each dot represents one idea's purchase intent rating averaged acros
mean rating. Ideas above the horizontal dashed line are the top 10% ideas. F
scale per Jamieson and Bass (1989).

Figure 3. 2D Projection of 512-Dimensional Human (Left) and AI (Right) Ideas



Notes. Figure shows 200 human ideas (left) and 200 AI ideas (right, both zero-shot and few-shot). The ideas are projected from their original USE-based embedding (512-dimensions) down to two-dimensional UMAP space. In this figure, the dimensions are shared between both conditions by embedding and projecting down all 400 ideas at once. In all statistical analyses, the pools are analyzed separately.

AI and Its Impact on Creativity and Diver
Empirical Study of LLM-Generated Prod

Cognitive bias risks

- **Confirmation bias**: not questioning the research because it confirms our beliefs
- **Illusory correlation**: seeing relationships that aren't proven
- **Negativity bias**: focusing only on risks
- **Availability heuristic**: using something because it's there
- **Status quo bias**: "if it ain't broke..."
- **Uniqueness bias**: we think we are more unique than we are

For now, I don't think there's strong evidence for AI-induced job losses, at least outside of certain contract types (freelancers) or roles (junior programmers). That's not to say it's a great time to be looking for a job, especially if you're young. But correlation is not causation. **There are many other factors impacting labour markets, from trade wars to the end of the era of cheap money**. On that note, it's worth raising an eyebrow when companies announcing lay-offs link them in vague terms to becoming "AI ready", but without any corroborating detail. Let's face it, it sounds a lot more dynamic than just saying the business isn't doing very well.

It's not black & white

John Burn-Murdoch and Sarah O'Connor, The AI Shift: where are all the job losses? FT, October 23 2025, https://www.ft.com/content/3d2669e3-c05e-48c9-8bb3-893c1d66de2e Sources: Who Is AI Replacing? The Impact of Generative AI on Online Freelancing Platforms (Demirci et al, 2024) and Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence (Brynjolfsson et al, 2025)

Every time jobs automate or we get mechanisation or computerisation, jobs disappear [and] people think there are not going to be any jobs. And every time so far, that was false. One of the reasons is: it's really hard to imagine new things coming up. That's a fundamental imagination problem, as to why we may be too pessimistic about technology sometimes. Daron Acemoglu and Pascual Restrepo have a paper where they found that **half of employment growth between 1980 and 2007 took place in occupations with new job titles. There were entirely new occupations that, for sure, people in 1970 and 1980 wouldn't have been able to imagine.**

It's not new

https://pinboard.in/cached/b8d26bf54cb0/

AI resembles many other aspects of our complex-socially mediated world in that it can never be fully explainable or transparent. Therefore, **rather than explaining or unpacking 'black boxes'—things that we do not understand and cannot explain—our graduates must learn to work and make judgements within complex organisational and social structures, where they need to synthesise information in a sophisticated way as part of their professional practices**.

It's not just AI

Bearman, M., & Ajjawi, R. (2023). Learning to work with the black box: Pedagogy for a world with artificial intelligence. British Journal of Educational Technology, 54, 1160–1173. https://doi.org/10.1111/bjet.13337

What are the opportunities?

1. Counter-balance human bias?
2. Improve human accuracy?
3. Reduce a project's overall environmental impact?
4. Improve security and privacy?
5. Empower? Accessibility, usability and reach
6. Free up time for more critical tasks?
7. Financial viability?

AI risks and opportunities assessment template

Use this form to systematically assess the risks associated with the use or non-use of AI for a particular task. Describe the task and its objective(s) first here: _____

Risk	**Examples** (human and AI)	**Risk assessment of BOTH human and AI** (including high/medium/low)	**Opportunities** (e.g. improve skills/ accuracy, reduce bias/energy, etc)
Accuracy	AI hallucinations, human error		
Objectivity, bias, fairness	Demographic, algorithmic, temporal, data-based. Positivity bias, verbosity, greediness. Human cognitive biases: confirmation bias, algorithm aversion.		
Environmental	What is the environmental impact of all activities associated with the task, factoring in time spent		
Employment & skills	Deskilling (e.g. cognitive offloading), potential for job displacement, redundancy or creation. Job satisfaction		
Time	Time-saving, time-adding		
Security, privacy, copyright	Sensitive material being used in training/seen by trainers/developers		

Then identify strategies

1. Risk reduction for critical thinking: trust vs skills
2. Risk reduction for accuracy: shift emphasis from gathering to verification/evaluation
3. Risk reduction for cheating: shift emphasis from execution to stewardship
4. Risk reduction for bias: emotional reflexivity

Lower trust, build skills

- “Professional **experience** and domain **expertise** are the most protective [against automation bias]”
- “Advanced **AI background knowledge** [calibrates] trust”
- “**Verification**-related cognitive engagement” helps debias
- **Anchoring bias** matters: a negative first AI experience
- Warn about **Dunning-Kruger Effect** (some confidence)

Romeo & Conti. Exploring automation bias in human–AI collaboration: a review and implications for explainable AI
https://link.springer.com/article/10.1007/s00146-025-02422-7

For recall and comprehension, ***the focus shifts from information gathering to information verification****.*

For application, the emphasis shifts ***from problem-solving to AI response integration****.*

For analysis, synthesis, and evaluation, ***effort shifts from task execution to task stewardship****.*

“Some cognitive tasks become less necessary due to GenAI. For instance, information gathering has been significantly reduced. GenAI tools automate the process of fetching and curating task-relevant information, making it less effortful for knowledge workers.”

The Impact of Generative AI on Critical Thinking: Self-Reported Reductions in Cognitive Effort and Confidence Effects From a Survey of Knowledge Workers - Microsoft Research

Emotional reflexivity:

"Emotional reflexivity is described by Olson et al. (2021, p. 2) as 'a process of drawing on emotions to, potentially, chart a unique path', and through this acknowledgement, understanding how emotions are shaping action and experience. Through considering—and even mobilising—feelings, people can navigate emotionally and relationally through the world. This is a useful framing. Both trust and distrust are powerful affective prompts that should not be overlooked by students—but nor are they sufficient in themselves. **Highlighting the need to be emotionally reflexive can prompt students to examine their interactions with AI systems in a critical way**." Bearman & Ajjawi

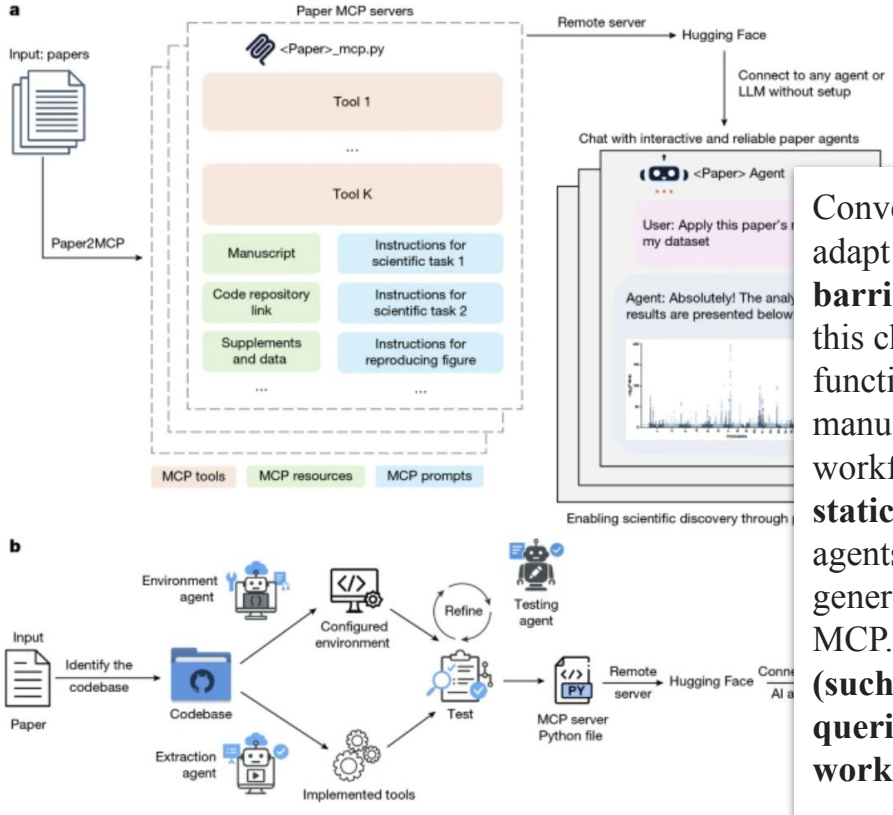
Emotional reflexivity:

"Emotional reflexivity is described by Olson et al. (2021, p. 2) as 'a process of drawing on emotions to, potentially, chart a unique path', and through this acknowledgement, understanding how emotions are shaping action and experience. Through considering—and even mobilising—feelings, people can navigate emotionally and relationally through the world. This is a useful framing. Both trust and distrust are powerful affective prompts that should not be overlooked by students—but nor are they sufficient in themselves. **Highlighting the need to be emotionally reflexive can prompt students to examine their interactions with AI systems in a critical way**." Bearman & Ajjawi

Negativity bias: not just risks

1. Opportunity to reduce human bias: persona-led design
2. Opportunity to empower: papers as agents
3. Opportunity to reduce human inaccuracy: roleplay prompting
4. Opportunity to upskill: journey prompting

Fig. 1: Overview of Paper2Agent.



Conventional research papers require readers to understand and adapt the paper's code, data and methods to their work, creating **barriers to dissemination and reuse**. Paper2Agent addresses this challenge by converting a paper into an AI agent that functions as a virtual corresponding author, exposing its manuscript, supplementary materials, datasets, code and workflows as **active, agent-native knowledge rather than static text**. It analyses the paper and codebase using multiple agents to construct a model context protocol (MCP) server, then generates and runs tests to refine and increase robustness of the MCP. These paper MCPs **can be connected to a chat agent (such as Claude Code) to carry out complex scientific queries through natural language while invoking tools and workflows from the paper**.

Miao, J., Davis, J.R., Zhang, Y. et al. Reimagining research papers as interactive and reliable AI agents. Nature (2026). https://doi.org/10.1038/s41586-026-11044-y

b

Digitised Persona	Emergent Archetype
 **Reggie** **Interests:** YouTubing, hanging out with friends. **Issues:** Talks too much in class, rather spend time with friends, family applies a lot of pressure on student – older sister that performs far better at school. **Engaging:** Enjoys 'hands-on' activities. Consider himself a bit of a joker, likes to be the centre of attention.	A student under pressure to perform, who focuses m… classmates tha… explanation fo… provided, alon… this potentially…
 **Thorin** **Interests:** Jazz music, playing computer games, reading, sleeping and talking. **Issues:** Sleeps in class, talks in class when tired, leaves others to help do his work, low digital technology skills. **Engaging:** Things relating to his interests, food, teacher attention. **Motivation:** to please, to receive praise, to please his mum.	A student who… due to a preoc… hobbies. His sc… result of priori… indicating chall… identifies attri… be used to eng…
 **Maxton** **Interests:** Friendships, music, hanging out at school with his friends.	A student who prioritises socialising and other activities above school work. Despite this, the student behaves in class and follows direction, though his attention may be elsewhere.

A three-step activity framework for using creative personas effectively in similar situations where a group of teachers-as-designers are co-designing for absent students. It proposes that **personas are valuable for groups of teachers co-designing for students as a way to evoke empathy, aid in problem framing, and set up a space for creative thinking** as part of an iterative design process.

Kerr, J., Kelly, N. Use of personas in co-designing learning experiences with teachers: An exploratory case study. Int J Technol Des Educ 35, 189–207 (2025). https://doi.org/10.1007/s10798-024-09900-x

4 role play design techniques for genAI



Rubber ducking

Using AI for 'self explanation' to work through a problem.



Red teaming/ devil's advocate

Using AI to identify potential lines of attack by an adversary, or potential flaws/gaps in a story.



Critical friend/mentor

Using AI for feedback or guidance while avoiding deskilling.



Audience personas

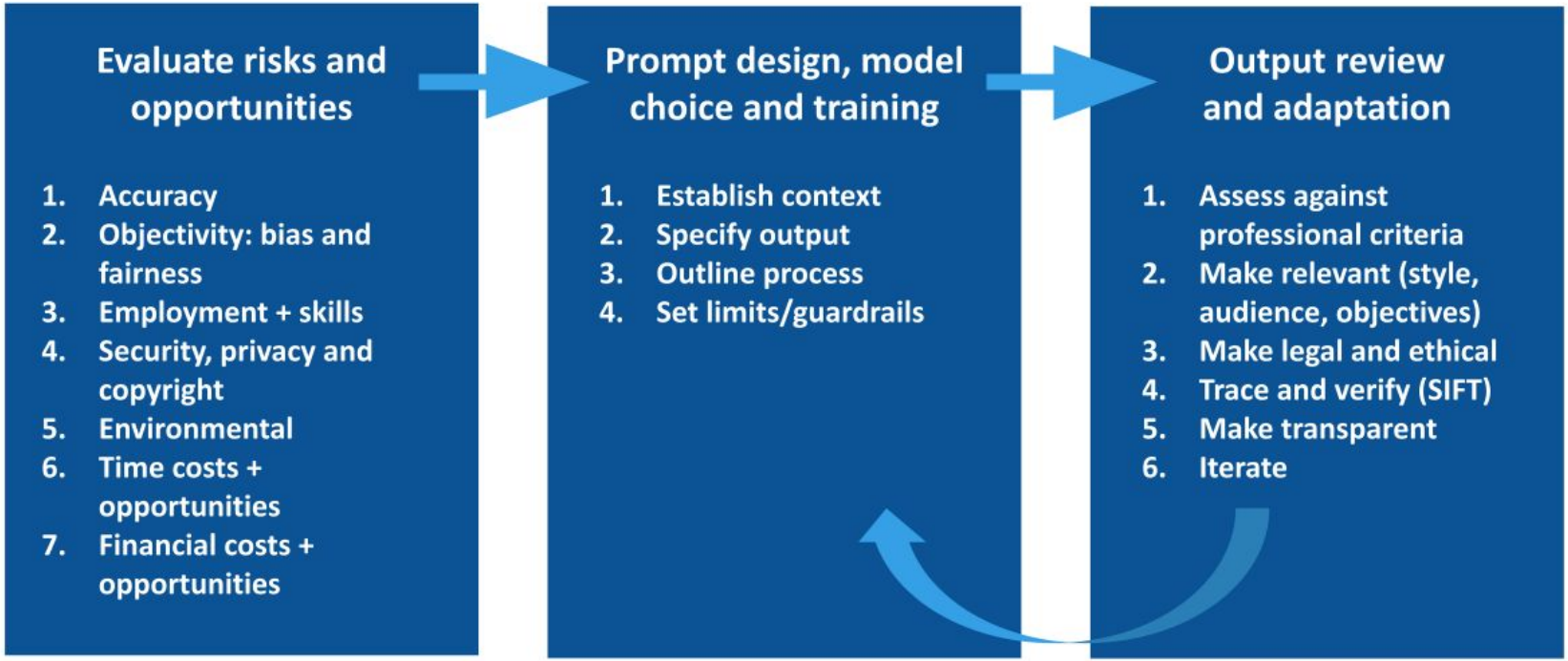
Using AI to review content from the position of the target audience.

ONLINE JOURNALISM BLOG

Paul Bradshaw 2025 | Icons generated by Google Gemini

[4 ways you can 'role play' with AI]

Three steps of professional genAI use



Training might also include harnesses, models, collecting training data, optimisation

Paul Bradshaw 2026

For a project/assessment...

List risks and opportunities related to

- Accuracy
- Objectivity/bias/fairness
- Employment/skills
- Environment
- Security/privacy
- Time/money